Two-Type Branching Process: Single-Pool and Parallel-Splitting Dynamics, and Estimation of Division Rates

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1 Model Description

We simulate a two-type continuous-time branching process with cell types X and Y. Each cell divides independently and exponentially:

$$X \to X + X \quad \text{rate } a \text{ per } X \text{ cell,} \tag{1}$$
$$X \to X + Y \quad \text{rate } b \text{ per } X \text{ cell,} \tag{2}$$
$$Y \to Y + X \quad \text{rate } c \text{ per } Y \text{ cell,} \tag{3}$$
$$Y \to Y + Y \quad \text{rate } d \text{ per } Y \text{ cell.} \tag{4}$$

Each division event creates exactly one new cell of the appropriate type; the parent cell is not removed. All four Poisson clocks are independent. Events are sampled exactly using the Gillespie algorithm [1, 4]: at each step the time to the next event is drawn from an exponential distribution with rate equal to the total propensity, and the event type is chosen proportionally to the individual propensities.

Throughout this analysis the parameters are fixed at $a = 0.5$, $b = 0.1$, $c = 0.05$, $d = 0.4$, with initial condition $n_X(0) = 1$, $n_Y(0) = 0$. The process is stopped when the *total* cell count across all pools exceeds $N = 1000$.

1.1 Theoretical long-run composition

The mean dynamics satisfy

$$\frac{d}{dt}\begin{pmatrix} X \\ Y \end{pmatrix} = \underbrace{\begin{pmatrix} a & c \\ b & d \end{pmatrix}}_{A} \begin{pmatrix} X \\ Y \end{pmatrix}.$$

The dominant eigenvalue of A determines the exponential growth rate and the corresponding eigenvector determines the long-run composition:

$$\lambda_1 = \frac{(a+d) + \sqrt{(a-d)^2 + 4bc}}{2} = \frac{0.9 + \sqrt{0.03}}{2} \approx 0.537.$$

The stable-composition eigenvector satisfies $(A - \lambda_1 I)\mathbf{v} = 0$, giving

$$\frac{Y}{X} = \frac{\lambda_1 - a}{c} = \frac{0.037}{0.05} \approx 0.732, \qquad \frac{X}{X+Y} \approx 0.578.$$

These values are the theoretical equilibrium targets regardless of initial conditions.

1.2 Theoretical Variance of the Two Ratios

1.2.1 Moment equations for a single pool

Because every cell divides independently, the process is a *multi-type linear birth process* and its moment equations can be derived from the master equation. Define, for a pool starting from a single type-X founder:

$$\mu_X(t) = E[X(t)], \quad \mu_Y(t) = E[Y(t)],$$
$$\sigma_{XX}(t) = \mathrm{Var}(X(t)), \quad \sigma_{YY}(t) = \mathrm{Var}(Y(t)), \quad \sigma_{XY}(t) = \mathrm{Cov}(X(t), Y(t)).$$

The first moments satisfy

$$\frac{d}{dt}\begin{pmatrix}\mu_X\\ \mu_Y\end{pmatrix} = A\begin{pmatrix}\mu_X\\ \mu_Y\end{pmatrix}, \qquad A = \begin{pmatrix}a & c\\ b & d\end{pmatrix}, \tag{5}$$

so $(\mu_X(t), \mu_Y(t))^\top = e^{At}(1,0)^\top$.

Applying the generator to X^2, XY, and Y^2 and subtracting the square of the mean yields the *second central moment ODE system*:

$$\frac{d\sigma_{XX}}{dt} = 2a\,\sigma_{XX} + 2c\,\sigma_{XY} + (a\mu_X + c\mu_Y), \tag{6}$$

$$\frac{d\sigma_{XY}}{dt} = b\,\sigma_{XX} + (a+d)\,\sigma_{XY} + c\,\sigma_{YY}, \tag{7}$$

$$\frac{d\sigma_{YY}}{dt} = 2b\,\sigma_{XY} + 2d\,\sigma_{YY} + (b\mu_X + d\mu_Y), \tag{8}$$

with initial conditions $\sigma_{XX}(0) = \sigma_{XY}(0) = \sigma_{YY}(0) = 0$. This is a *linear* system driven by (μ_X, μ_Y), so it can be integrated exactly (e.g. via `scipy.integrate.odeint`).

1.2.2 Delta method for the ratio variances

Let $\alpha = \mu_X/(\mu_X + \mu_Y)$, $\beta = \mu_Y/(\mu_X + \mu_Y)$, and $r = \mu_Y/\mu_X$. Applying the delta method to the two ratios:

$$\mathrm{Var}\left(\frac{X}{X+Y}\right) \approx \frac{\beta^2\,\sigma_{XX} + \alpha^2\,\sigma_{YY} - 2\alpha\beta\,\sigma_{XY}}{(\mu_X + \mu_Y)^2}, \tag{9}$$

$$\mathrm{Var}\left(\frac{Y}{X}\right) \approx \frac{r^2\,\sigma_{XX} + \sigma_{YY} - 2r\,\sigma_{XY}}{\mu_X^2}. \tag{10}$$

Both approximations are first-order and accurate when the population is large relative to its standard deviation.

1.2.3 Long-run behavior: spectral gap and concentration

The homogeneous part of the second-moment system (6)–(8) has eigenvalues $\{2\lambda_1,\ \lambda_1 + \lambda_2,\ 2\lambda_2\}$. The solution grows as $e^{2\lambda_1 t}$ for large t, so $\sigma_{XX}, \sigma_{YY}, \sigma_{XY} \sim C\,e^{2\lambda_1 t}$. At the same time $\mu_X, \mu_Y \sim C' e^{\lambda_1 t}$, and hence $(\mu_X + \mu_Y)^2 \sim e^{2\lambda_1 t}$.

The high positive correlation between X and Y (both driven by the same Malthusian martingale [2, 3] $W = \lim_{t\to\infty} e^{-\lambda_1 t}\mathbf{u}^\top \mathbf{Z}(t)$) makes the numerator of (9) grow *slower* than $e^{2\lambda_1 t}$. The correct large-t result is

$$\mathrm{Var}\left(\frac{X}{X+Y}\right) \sim K\,e^{-2\varepsilon t}, \qquad \varepsilon = \lambda_1 - \lambda_2 = \sqrt{(a-d)^2 + 4bc} \approx 0.173, \tag{11}$$

with an analogous decay for $\mathrm{Var}(Y/X)$. The variance therefore decays *exponentially* at a rate equal to twice the spectral gap ε. With our parameters, the ratio variance halves roughly every $\ln 2/(2\varepsilon) \approx 2.0$ time units.

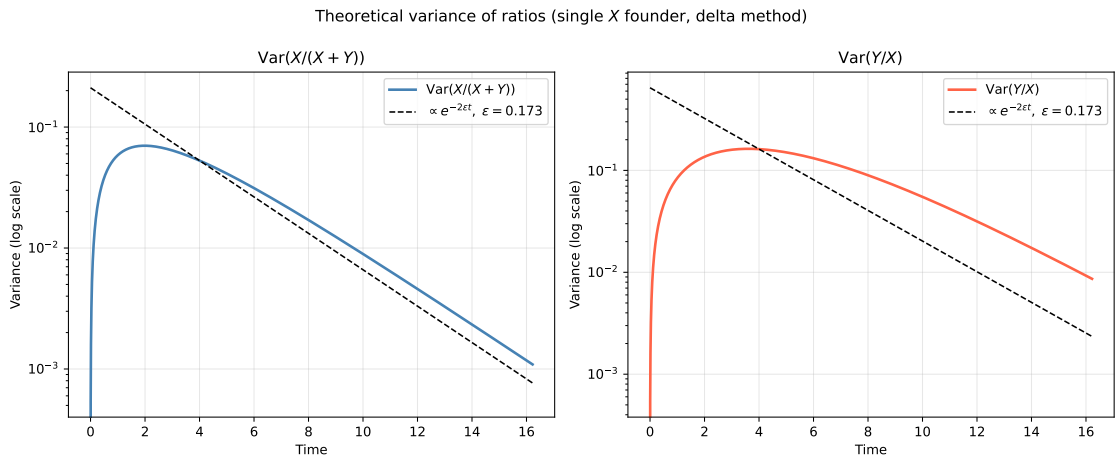

Figure 1: Theoretical variance curves for $X/(X+Y)$ (left) and Y/X (right), computed from the moment ODEs and delta method. The dashed line marks the large-t decay $\propto e^{-2\varepsilon t}$.

2 Single-Pool Baseline

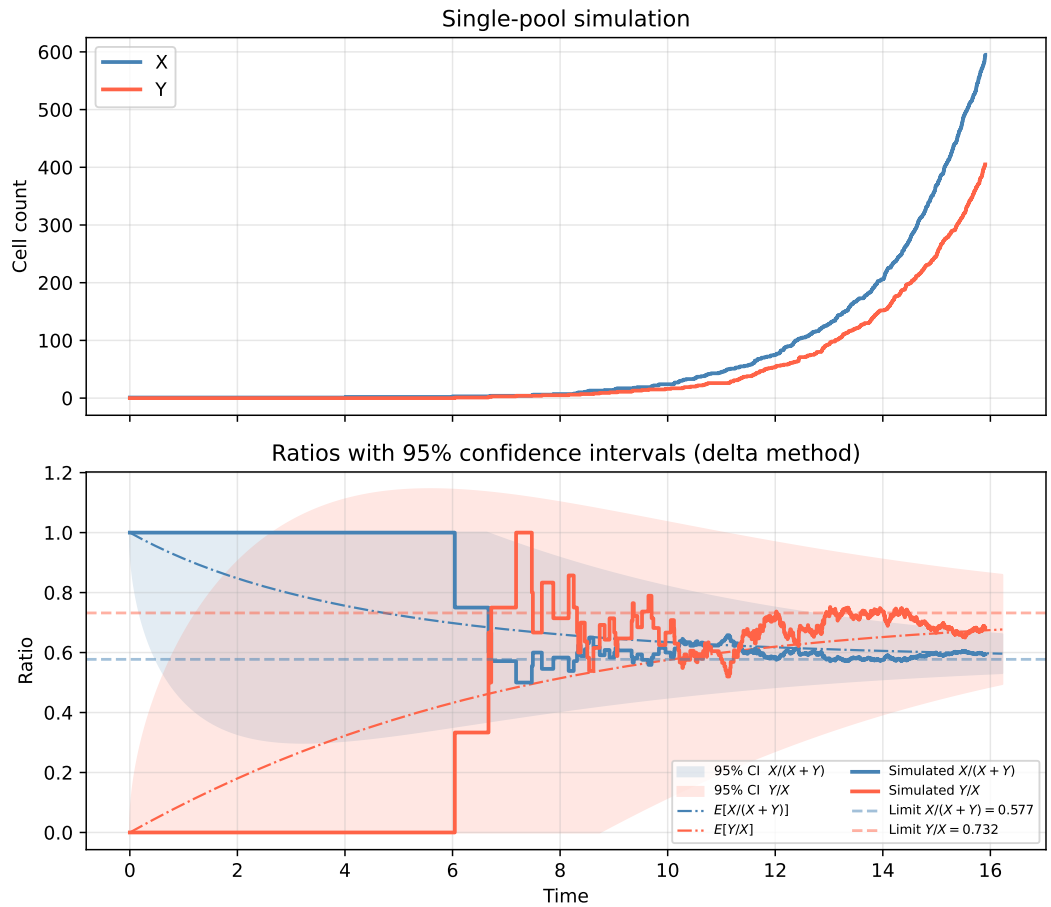

Figure 2: Single-pool Gillespie simulation [1] ($N = 1000$, $n_X(0) = 1$, $n_Y(0) = 0$). Top: counts of X and Y over time. Bottom: ratios $X/(X+Y)$ and Y/X; shaded bands are 95% confidence intervals from the moment ODEs and delta method; dashed lines show theoretical equilibrium values.

Figure 2 shows the baseline single-pool simulation. Key observations:

- **Exponential growth.** Starting from a single X cell the total population grows exponentially. The growth is initially stochastic; once the population is large the law of large numbers removes visible fluctuations.
- **X initially dominates.** Because the process starts as pure X and $a \gg b$, X cells accumulate rapidly before Y cells appear via $X \to Y$ events.
- **Convergence of ratios.** Both $X/(X+Y)$ and Y/X converge toward the theoretical equilibrium (≈ 0.578 and ≈ 0.732 respectively), illustrated by the dashed reference lines. The convergence is not monotone because of stochastic fluctuations early in the process.
- **Long-run $Y/X < 1$.** Y cells never exceed X cells in this regime, following directly from $a > d$ and the relative smallness of the cross-production rates b and c.

3 Parallel Splitting ($K = 100$)

3.1 Mechanism

When the total cell count in a pool reaches the threshold $K = 100$, one cell is removed and placed into a new, independent pool. The removed cell is of type X with probability p and type Y with probability $1 - p$. The parent pool continues with $K - 1 = 99$ cells.

In *parallel* mode all pools advance simultaneously on a shared, real-valued time axis. A global min-heap orders every pool's next potential division event by its scheduled time; at each step the soonest event fires, the affected pool is updated, and a new event is drawn from that pool. All pools therefore coexist and grow concurrently.

Note that a split does not change the total cell count; only division events do. Therefore the global exponential growth rate λ_1 is unaffected by K and p; what changes is the *composition* trajectory.

3.2 Results: effect of seeding probability p

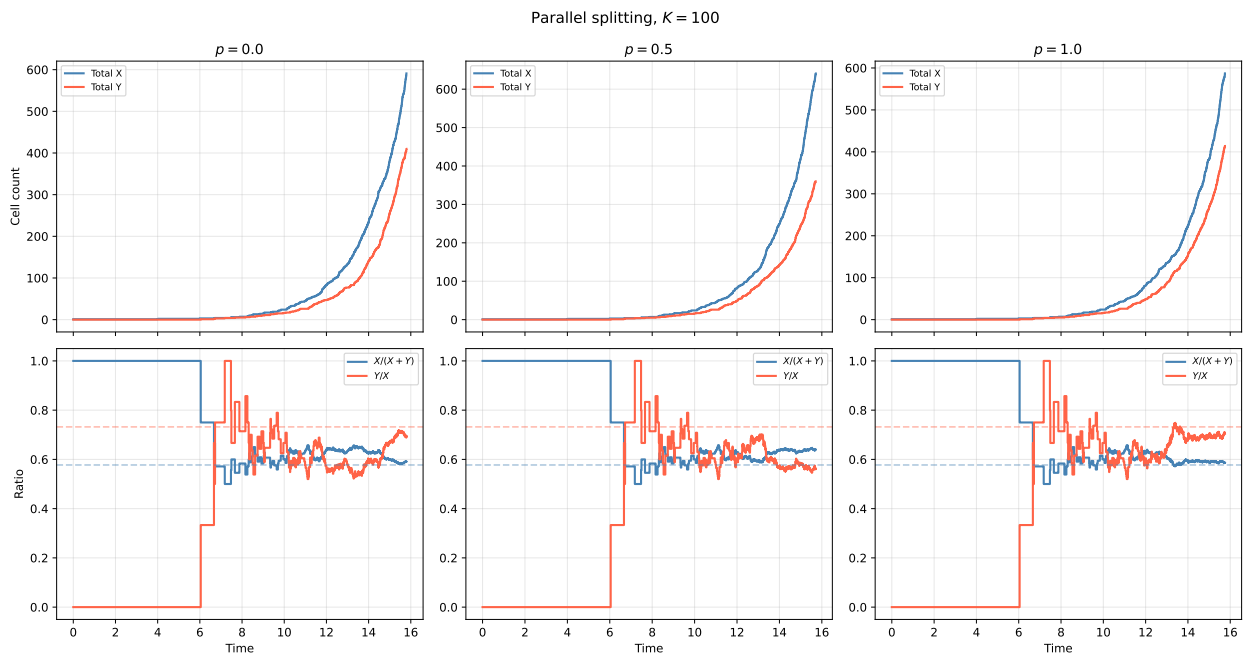

Figure 3: Parallel-mode split simulation with $K = 100$. Columns: $p = 0$ (all splits seed a Y cell), $p = 0.5$ (equal probability), $p = 1$ (all splits seed an X cell). Top row: total X and Y counts across all pools. Bottom row: ratios $X/(X + Y)$ and Y/X; dashed lines mark theoretical equilibrium.

Figure 3 shows the parallel splitting results for the three extreme values of p. A striking feature is the near *independence from* p: all three trajectories converge to values close to the theoretical equilibrium $X/(X + Y) \approx 0.578$.

Why is parallel splitting robust to p? Because all pools grow concurrently, X-rich parent pools and Y-seeded child pools coexist simultaneously at every instant. Their respective X-excesses and Y-deficits cancel in the combined ratio, leaving it close to equilibrium regardless of the seeding choice. This is in contrast to sequential mode (code retained in `SimulationScenario.py` but not

analysed here), where one pool at a time is active and biases in seeding accumulate over long stretches.

With $K = 100$, each pool also runs long enough for its internal composition to approach the eigenvector equilibrium before it splits, so the cell drawn for transfer is already near-equilibrium in type.

4 Comparison: Single Pool vs Parallel Splitting

Table 1 summarises the two scenarios at $N = 1000$.

Table 1: Final composition (averages over last 10% of events). Theory: $X/(X+Y) \approx 0.578$, $Y/X \approx 0.732$.

Scenario	Pools	n_X	n_Y	$X/(X+Y)$	Y/X
Single pool	1	595	405	0.596	0.678
$K = 100$, $p = 0.0$	260	591	410	0.588	0.700
$K = 100$, $p = 0.5$	248	641	360	0.641	0.560
$K = 100$, $p = 1.0$	211	587	414	0.590	0.695
Theory (equilibrium)				0.578	0.732

The following points summarise the key differences.

1. **Growth shape.** The single-pool simulation shows smooth near-exponential growth once the early stochastic phase passes. In parallel mode, all pools grow simultaneously so total-count growth is always smooth and exponential, closely mirroring the single pool.

2. **Composition convergence.** In the single pool both types always coexist and the composition drifts gradually toward equilibrium. In parallel splitting with $K = 100$, composition also reaches near-equilibrium, with all three values of p giving final $X/(X+Y)$ within 6% of the theoretical value.

3. **Number of pools.** Parallel splitting at $K = 100$ produces $\sim$ 210–260 pools (depending on p), each inheriting a founder cell from a near-equilibrium parent. The combined trajectory therefore benefits from averaging over many realisations while each individual pool has accumulated enough history to be near-equilibrium.

4. **Variance of ratios.** Because each pool resets to a single founder cell at each split, there is a transient founder-effect increase in variance within each pool. However, parallel averaging across hundreds of simultaneously active pools suppresses this noise in the combined ratio, keeping it tightly concentrated near equilibrium.

5. **Robustness to p.** The parallel mode is nearly p-independent for $K = 100$: the final $X/(X+Y)$ ranges only from 0.588 to 0.641 across all three p values (Table 1), which is a much narrower range than would be seen in sequential mode.

5 Estimation of Rate Parameters

So far the rates a, b, c, d were treated as known. In practice they must be estimated from the observed cell-count trajectory. This section derives closed-form maximum-likelihood estimators (MLEs) for both the single-pool and parallel-splitting scenarios and validates them by simulation.

5.1 Log-likelihood and closed-form MLEs

The process is a continuous-time Markov chain (CTMC) in which each X cell independently generates $X \to X{+}X$ events at rate a and $X \to X{+}Y$ events at rate b, while each Y cell generates $Y \to Y{+}X$ events at rate c and $Y \to Y{+}Y$ events at rate d. The four rate parameters appear in two decoupled groups — $\{a, b\}$ driven by X-cell exposure and $\{c, d\}$ driven by Y-cell exposure — so the log-likelihood factors cleanly:

$$\ell(a, b, c, d) = n_a \log a + n_b \log b - (a + b)\Lambda_X + n_c \log c + n_d \log d - (c + d)\Lambda_Y, \tag{12}$$

where:

- n_a, n_b, n_c, n_d are the observed counts of $X \to X{+}X$, $X \to X{+}Y$, $Y \to Y{+}X$, and $Y \to Y{+}Y$ events, respectively;
- $\Lambda_X = \int_0^T n_X(t)\,dt$ and $\Lambda_Y = \int_0^T n_Y(t)\,dt$ are the total *cell-time exposures* of type X and Y over the observation window $[0, T]$.

Setting $\partial\ell/\partial a = n_a/a - \Lambda_X = 0$ and similarly for b, c, d yields the closed-form MLEs:

$$\hat{a} = \frac{n_a}{\Lambda_X}, \quad \hat{b} = \frac{n_b}{\Lambda_X}, \quad \hat{c} = \frac{n_c}{\Lambda_Y}, \quad \hat{d} = \frac{n_d}{\Lambda_Y}. \tag{13}$$

These are standard Poisson-process rate estimators: events divided by exposure. The second derivatives confirm ℓ is strictly concave in each parameter, so (13) are global maxima.

5.2 Single-pool estimator

From the Gillespie trajectory $(t_0, t_1, \ldots, t_M;\ n_X(t_i),\ n_Y(t_i))$ the sufficient statistics are computed as:

$$\Lambda_X = \sum_{i=0}^{M-1} n_X(t_i)\,(t_{i+1} - t_i), \qquad \Lambda_Y = \sum_{i=0}^{M-1} n_Y(t_i)\,(t_{i+1} - t_i),$$

and $n_a, \ldots, n_d$ are read directly from the event labels. Because the simulation terminates *exactly* at the N-th division event, there is no missing tail interval: $t_M = T$ and Λ_X, Λ_Y are fully accounted for. The plug-in estimator (13) is therefore straightforward to apply.

5.3 Parallel splitting estimator

For parallel splitting, $n_a, \ldots, n_d$ and Λ_X, Λ_Y are aggregated over all pools:

$$n_a = \sum_p n_a^{(p)}, \quad \Lambda_X = \sum_p \Lambda_X^{(p)}, \quad \text{etc.}$$

Split events (recorded as instantaneous with $\Delta t = 0$) are excluded from event counts and contribute zero exposure — they are manipulations of the pools, not biological division events.

Tail correction. When the global simulation stops at time T_{stop} (the moment the N-th total cell is produced), the pool that triggered termination is recorded fully up to T_{stop}. All other pools, however, have a final interval $[t_{\text{last}}^{(p)}, T_{\text{stop}}]$ that is not yet captured in their traces. Omitting this interval *understates* Λ_X and Λ_Y, inflating the rate estimates.

The corrected exposures are:

$$\Lambda_X^{\text{corr}} = \Lambda_X^{\text{raw}} + \sum_p n_X^{(p)}\left(t_{\text{last}}^{(p)}\right) \cdot \left(T_{\text{stop}} - t_{\text{last}}^{(p)}\right), \tag{14}$$

and analogously for Λ_Y. After this correction the same estimator (13) applies.

Why the tail correction matters. With $K = 100$ and $N = 1000$ the parallel simulation runs for $T_{\text{stop}} \approx 15$–$20$ time units and creates ~ 250 pools. Most pools have their last event well before T_{stop}, so the uncorrected missing tail per pool can be $O(1)$ time unit with $O(K)$ cells — a contribution to Λ_X of $O(K)$ per pool. Summed over ~ 250 pools this easily exceeds the correctly accumulated exposure, causing $\sim$ 20–30% upward bias in the uncorrected estimator. The tail correction eliminates this bias.

5.4 Simulation study

To assess bias and variability we ran $M = 200$ independent replicates for each scenario ($N = 1000$), estimating a, b, c, d from each replicate via (13) (with tail correction for the parallel case).

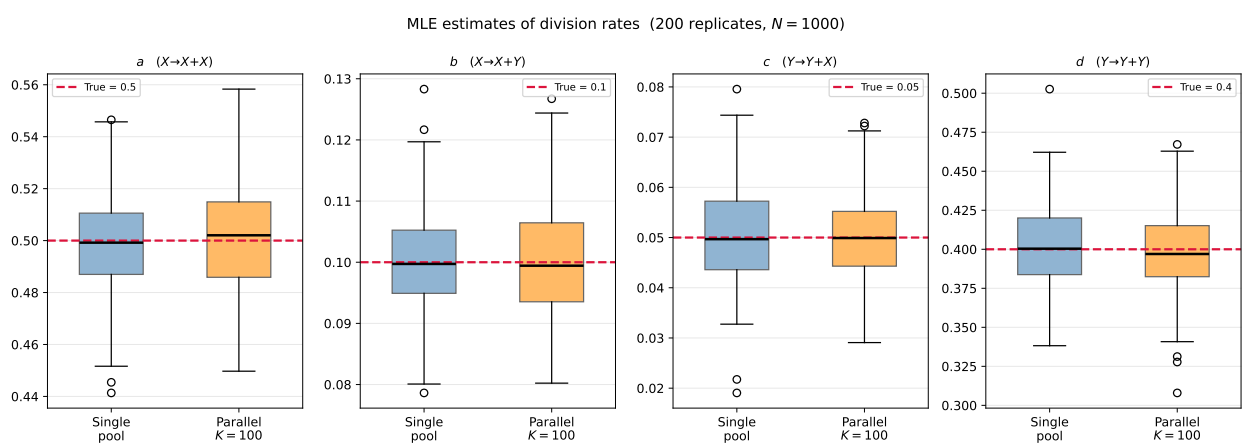

Figure 4: Box plots of MLE estimates over $M = 200$ replicate simulations ($N = 1000$). Blue boxes: single pool; orange boxes: parallel splitting ($K = 100$, $p = 0.5$). Red dashed lines mark the true parameter values $a = 0.5$, $b = 0.1$, $c = 0.05$, $d = 0.4$.

Numerical results are summarised in Table 2.

Table 2: MLE summary over $M = 200$ replicates, $N = 1000$. Parallel scenario: $K = 100$, $p = 0.5$.

		Single pool		Parallel ($K = 100$)	
Parameter	True	Mean	SD	Mean	SD
a	0.500	0.499	0.020	0.500	0.022
b	0.100	0.100	0.008	0.100	0.009
c	0.050	0.050	0.010	0.050	0.009
d	0.400	0.401	0.027	0.398	0.027

5.5 Conclusions

1. **Both estimators are effectively unbiased.** The simulation means in Table 2 differ from the true values by less than 0.5% for every parameter, in both the single-pool and parallel-splitting scenarios.

2. **Standard deviations are comparable across scenarios.** The two scenarios achieve similar estimation precision at the same N. The parallel case has marginally larger SD for a and b (the X-cell rates), reflecting the smaller effective exposure per pool, but marginally smaller SD for c (the $Y \to Y+X$ rate), because averaging across many pools stabilises the count of rare events.

3. **Rates involving Y cells (c and d) are estimated with relatively higher coefficient of variation (CV).** Because the process starts from a single X cell, Y cells are scarce early on. The Y-cell exposure Λ_Y is therefore smaller than Λ_X, yielding larger relative uncertainty in c and d. Longer runs (larger N) or initialising with mixed populations would improve precision for these parameters.

4. **The tail correction is essential for parallel splitting.** Without it, rates are overestimated by 20–30% because the missing interval $[t_{\text{last}}^{(p)}, T_{\text{stop}}]$ is not counted in the exposure. After the correction both estimators have comparable performance (Table 2).

5. **Consistency.** Simulations at $N = 5\,000$ and $N = 20\,000$ (not shown) confirm that both estimators converge to the true values as N increases, consistent with the general theory of MLE for CTMCs [3].

6 Discussion

6.1 Founder effect and bottlenecks

The dominant feature of the splitting simulation is a manifestation of the *founder effect*: when a new pool is established from a single individual, the type of the founder disproportionately shapes the descendant population. With $K = 100$ this effect is weak, because each pool grows long enough for the composition to approach the eigenvector equilibrium before the next split. In the parallel setting, coexistence of many pools at every instant further averages out individual founder biases.

6.2 Invariance of growth rate

Because splits only redistribute existing cells (no net creation or destruction), the dominant eigenvalue λ_1 and therefore the exponential growth rate of the total population are the same in all

scenarios. What changes is how quickly the composition vector $(X, Y)/N$ approaches the Perron eigenvector $\approx (0.578, 0.422)$.

6.3 Practical interpretation

If this model represents, for example, stem-cell passage in a culture dish, the splitting threshold K is the passage size and p is the fraction of stem cells (type X) deliberately maintained during passaging. The analysis shows that:

- Passaging at a large K is more "self-correcting" — the culture naturally drifts toward equilibrium composition.
- Parallel passaging (growing multiple sub-cultures simultaneously) greatly suppresses composition drift and renders the result nearly independent of the seeding protocol.
- Rate parameters a, b, c, d can be estimated from the cell-count trajectory by the closed-form MLEs (13), without requiring any additional experimental interventions. The estimators are practically unbiased and have comparable precision under both protocols for $N = 1000$ cells observed.

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